

CUSUM Change-Point Detection Analysis

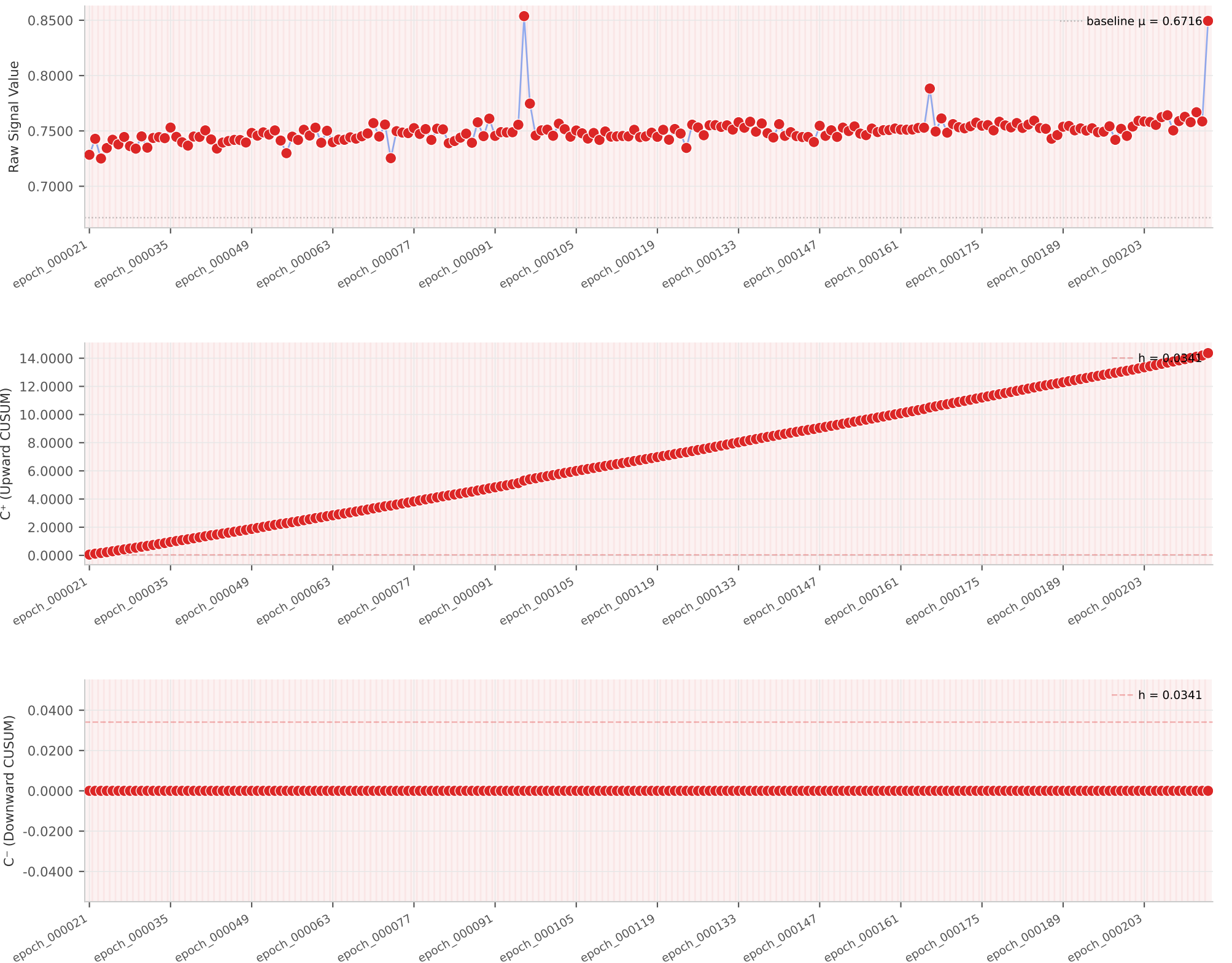
Cumulative Sum Control Chart | JS Divergence + New IP Ratio + Top-K Source

Run Parameters		Baseline CUSUM Parameters		Detection Summary (Any Signal)	
Dataset	epochs_10.0_2.0_cusum.json	Combined JS	$\mu=0.6716$ $\sigma=0.0085$ $k=0.0043$ $h=0.0341$	True Positives (TP)	194
Timestamp	2026-02-23T20:15:58.028325	New IP Ratio	$\mu=0.0000$ $\sigma=0.0000$ $k=0.0000$ $h=0.0000$	False Positives (FP)	0
Method	CUSUM Change-Point (JS + New IP + TopK)		$\mu=0.0623$ $\sigma=0.0024$ $k=0.0012$ $h=0.0096$	False Negatives (FN)	0
Total epochs	215			Precision	1.000
Baseline epochs	21 (10.0%)			Recall	1.000
Attack epochs	194			F1 Score	1.000
Malicious traffic	2.0%			Detection Rate	100.0%
CUSUM k-factor	0.5				
CUSUM h-factor	4.0				
Mode	FULL				

Combined JS Divergence

CUSUM Signal Analysis | malicious = 2.0% | baseline = 10.0%

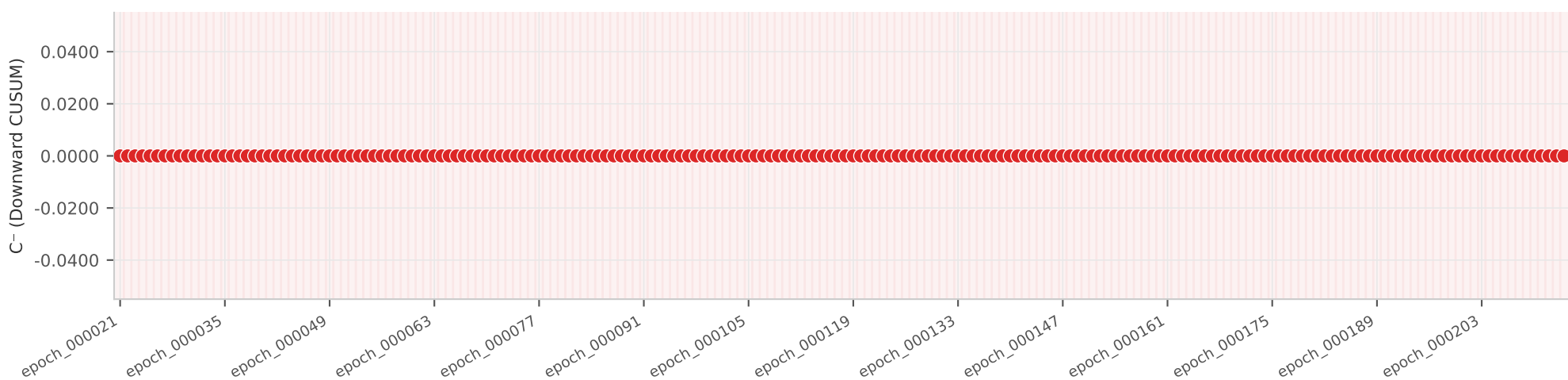
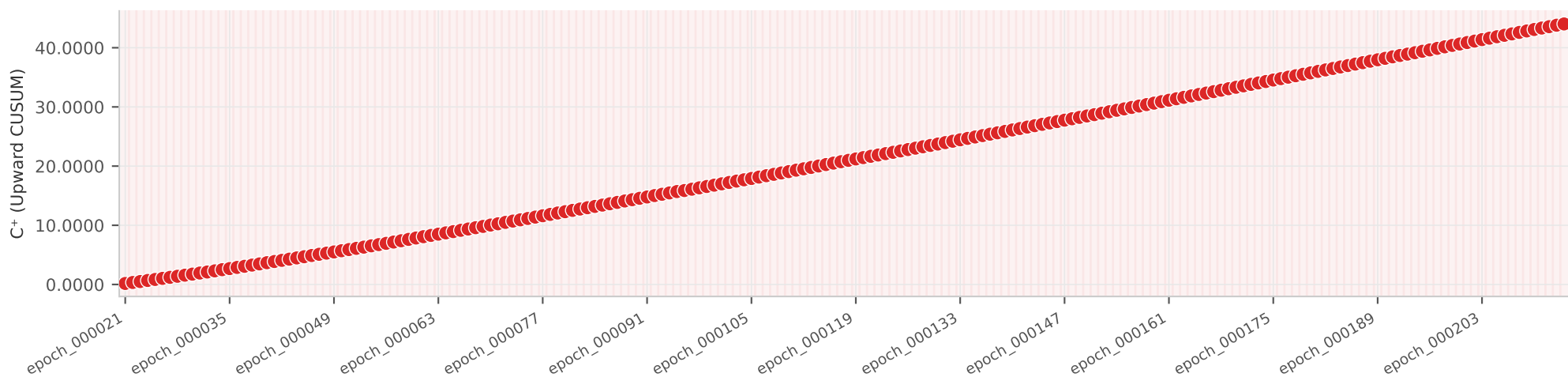
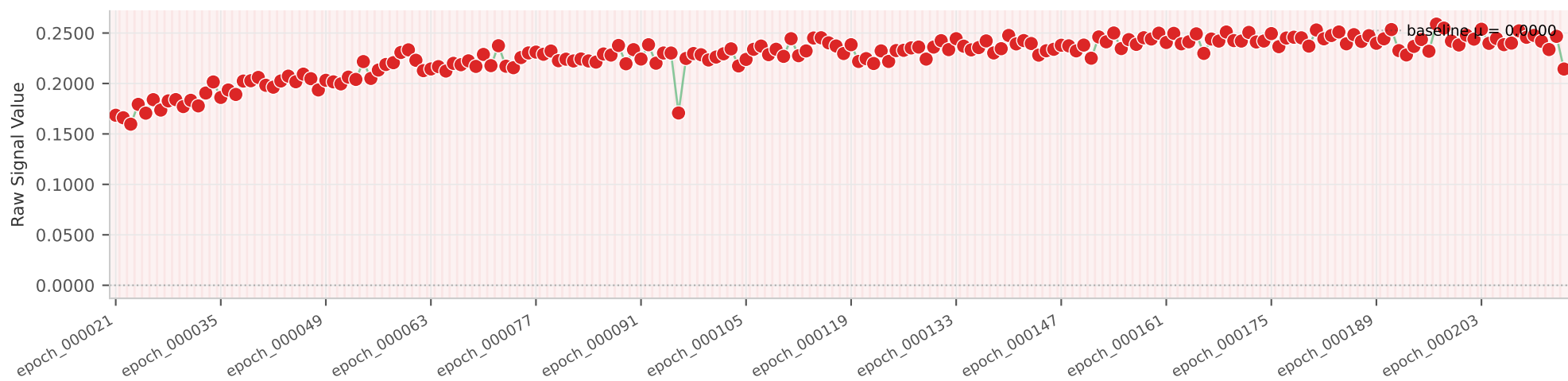
- CUSUM alarm fired
- Attacked epoch



New Source IP Ratio

CUSUM Signal Analysis | malicious = 2.0% | baseline = 10.0%

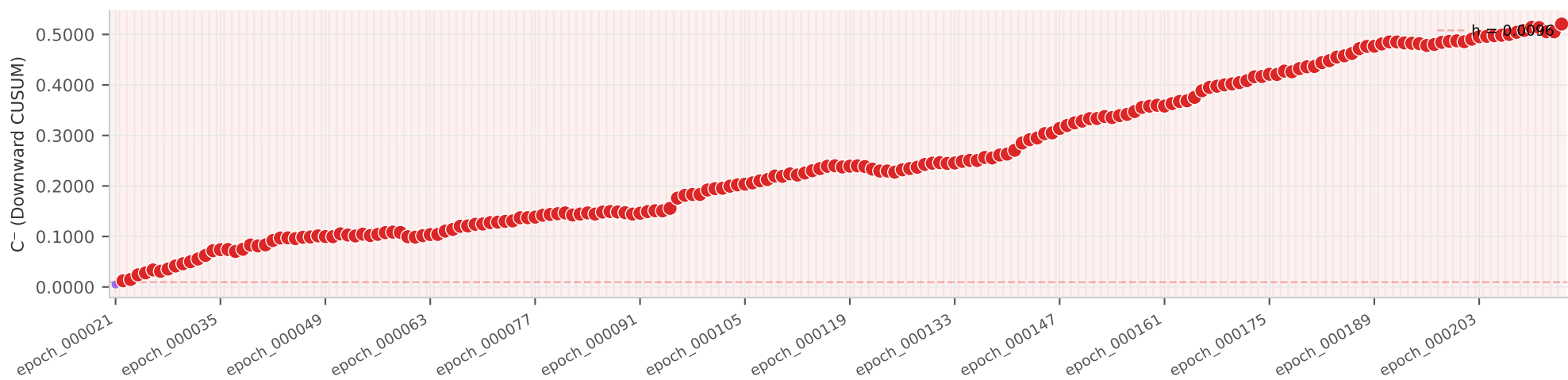
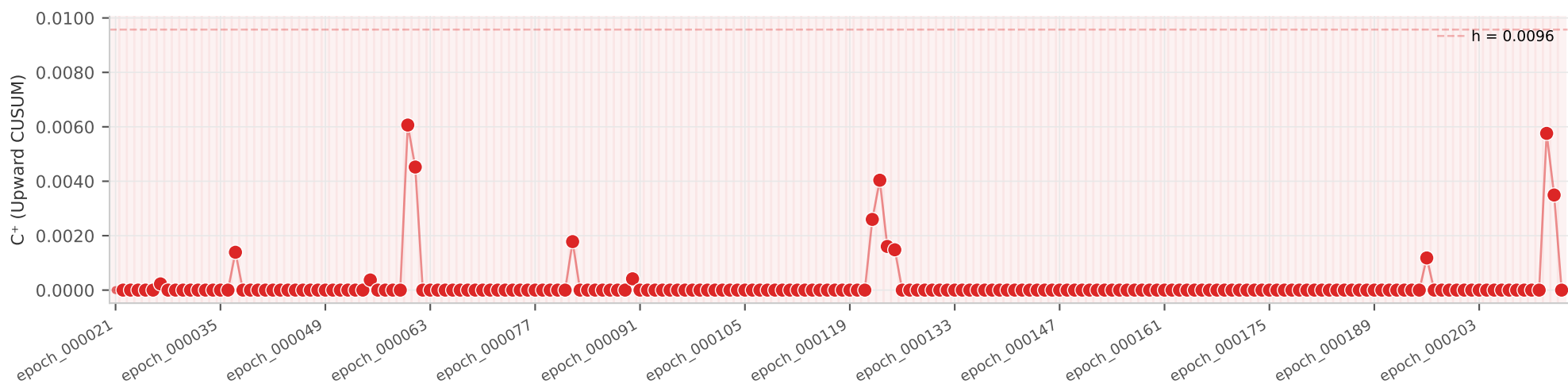
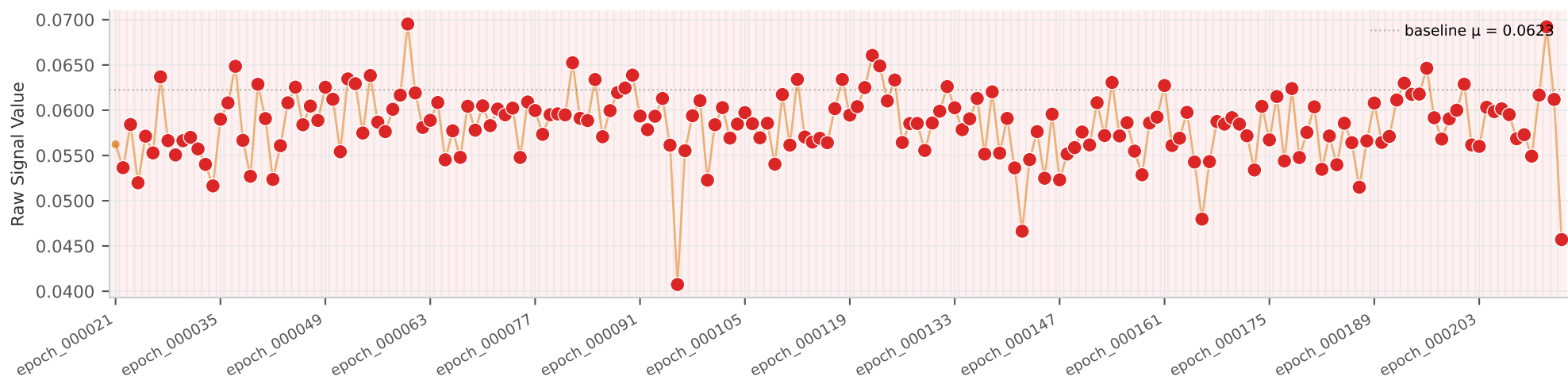
- CUSUM alarm fired
- Attacked epoch



Top-5 Source Concentration

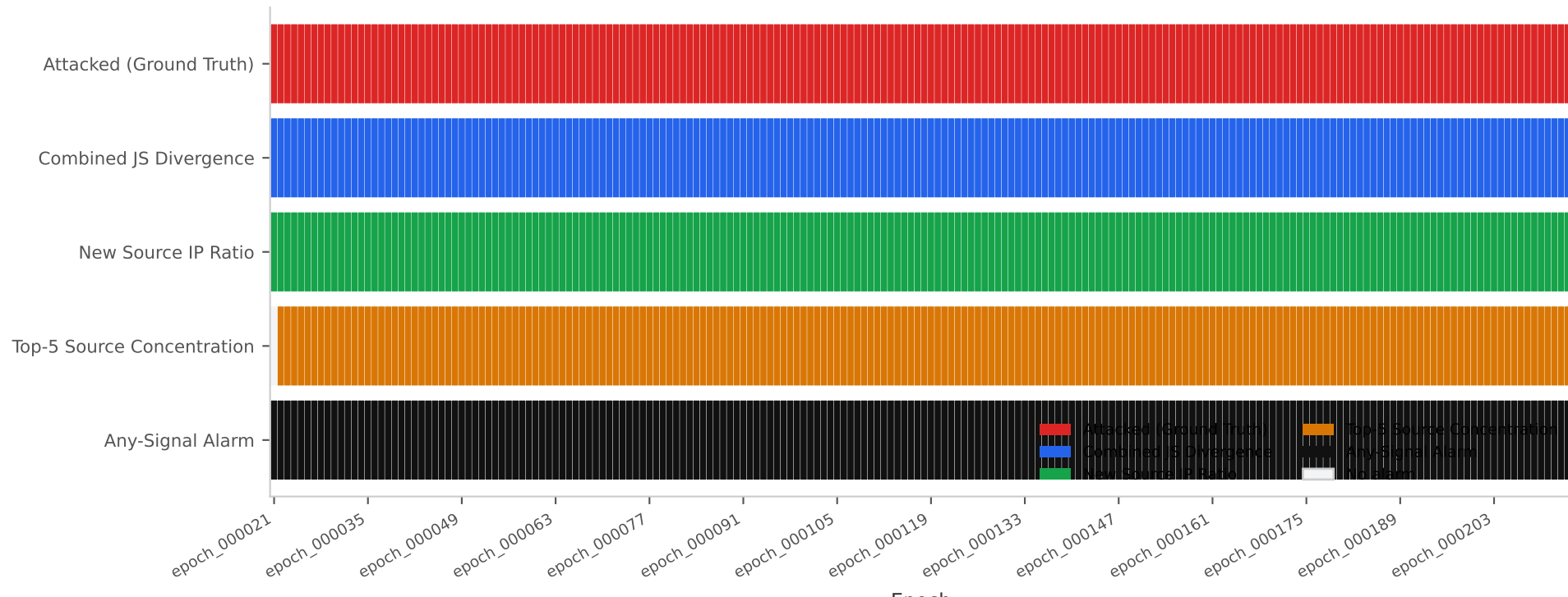
CUSUM Signal Analysis | malicious = 2.0% | baseline = 10.0%

- CUSUM alarm fired
- Attacked epoch



CUSUM Alarm Timeline

Filled cell = alarm fired for that epoch × signal



Detection Performance Metrics

Dataset: epochs | Malicious: 2.0% | Baseline: 10.0%

Metric	Value	Description
True Positives	194	Attacked epochs correctly alarmed
False Positives	0	Clean epochs incorrectly alarmed
True Negatives	0	Clean epochs correctly not alarmed
False Negatives	0	Attacked epochs missed
Precision	1.0000	$TP / (TP + FP)$
Recall	1.0000	$TP / (TP + FN)$
F1 Score	1.0000	Harmonic mean of Precision & Recall
Detection Rate	100.00%	Fraction of attacked epochs detected
False Alarm Rate	0.00%	Fraction of clean epochs with alarm